

Total No. of Questions : 8]

SEAT No. :

PB2341

[Total No. of Pages : 3

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B.E. (Information Technology)

BUSINESS ANALYTICS AND INTELLIGENCE

(2019 Pattern) (Semester - VIII) (414452C) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) List and explain the four processes that define a closed-loop BPM cycle.[8]
b) How is a balanced scorecard used in Business? [5]
c) What is data visualization? Explain the need for data visualization in Business intelligence. [5]

OR

- Q2)** a) List and briefly explain different types of charts used for visualization.[8]
b) List and briefly explain different types of Business Report. [5]
c) What is the role of the Performance dashboards? [5]

- Q3)** a) Write a short note on: [9]
i) Descriptive analytics
ii) Predictive analytics
iii) Prescriptive analytics
b) Consider a scenario that a bread seller charges Rs. 30 per loaf of bread, and he sells 50 loafs a day. What would happen to his revenue if he charged more for each loaf of bread? [8]

Evaluate the above mentioned scenario using what if analysis with respect to:

Case 1: The volume of bread sold doesn't depend on the price of the bread

Case 2: The amount of bread he sell does depend on the price he charge (Sale increases if price reduced, sale decreases if price increased)

Assume suitable data wherever required.

OR

P.T.O.

Q4) a) Discuss structure of mathematical model for decision support. **[8]**

b) Imagine you're managing the finances of a small business, and you want to plan your budget for the upcoming year. You have a basic income and expense sheet. You'd like to know how different changes will affect your end-of-year balance. **[9]**

Given the following information:

- Initial balance: \$10,000
- Monthly income: \$5,000
- Monthly expenses: \$3,000

You want to perform What-If Analysis for different scenarios:

- i) What would your end-of-year balance be if you increase your monthly income by 10%?
- ii) What if you cut your monthly expenses by 20%?
- iii) What if you do both, increase income by 10% and decrease expenses by 20%?

Q5) a) Discuss the role of predictive analytics in demand forecasting for production planning in a manufacturing unit. **[6]**

b) In logistics and production, what are the key performance indicators (KPIs) that organizations commonly track using BI? Explain the significance of these KPIs. **[6]**

c) Enlist the advantages of using Business Intelligence in Customer Relationship Management. **[6]**

OR

- Q6)** a) Compare Business Analytics (BA) with Business Intelligence (BI). [6]
- b) Explain application of Business Intelligence in Fraud Detection with a suitable example. [6]
- c) List various forms of Business Analytics and explain any one form in view of its support in practice. [6]
- Q7)** a) Write a short note on: Cloud Computing and Business Intelligence. [6]
- b) Enlist the factors that can be considered for implementing a Business Intelligence strategy? [5]
- c) In today's data-driven world, data analysis plays a crucial role in predicting future trends and outcomes. Discuss the key steps and considerations involved in using data analysis to make predictions about the future. Provide an example of a real-world application where predictive analysis has been successfully employed to make informed decisions. [6]

OR

- Q8)** a) Write a short note on: Issues of Legality, Privacy and Ethics. [6]
- b) You are the IT manager of a medium-sized company, and you are considering moving your business intelligence (BI) system to the cloud. Your BI system is critical for making data-driven decisions. You need to make a decision on which cloud service model (IaaS, PaaS, SaaS) to choose for hosting your BI system.
- Explain the key factors you should consider when making this decision and the potential advantages and disadvantages of each service model in the context of BI. [6]
- c) Write a short note on Analytics Applications for Consumers. [5]

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